The Theory of Closed Native Verification Systems

Version 4.0

This document constitutes a more rigorous formalization attempt of the CNVS Theory presented in the previous documentation.
Additional documents are expected to be released that will replace or supplement the present work.

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1. Introduction

This document constitutes the formalization of a theoretical system proposed as an Ideal Model for the verification attestation of physical evidence of the existence of an object external to the system itself, starting from declarations made within the system.

In this document, the term *Object* is intended in its broadest sense as any reality that can be supported by a specific, unique, and objectively identifiable physical observation.

In summary, this theoretical model attempts to reconcile a specific truth claim with the objective verification of its consistency within physical reality.

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2. Abstract

This document defines a new class of systems called **Closed Native Verification Systems (CNVS)**.

In such systems, the existence, validity, and evolution of state are determined exclusively by internal verification processes supported by externally generated

verification evidence, randomly distributed and ultimately governed by deterministic invariants.

Although objective verification processes remain external, their distribution—namely, the specific structure imposed by the system—reduces trust to the execution of a strict procedural framework that limits executor discretion through the progressive reduction of local and global operational knowledge.

The theory assumes three fundamental axioms that are imposed as necessary conditions for classifying a system as a Native Verification System.

The axioms define:

- the structural properties of the entire system;
- the emergent behavior related to security and verification dynamics;
- **the principles governing both external and internal processes.**

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3. Problem Statement

In traditional systems addressing the same problem:

- the existence of a state is a unilateral decision made by a governing authority (total trust);
- verification is entrusted to external auditors, generally constrained only by global rules with limited local references;
- the existence of state is admitted natively through issuer authority;
- state verification follows the declaration of existence;
- state is typically unitary;
- internal security decreases as adoption and the number of states increase, because the attack surface expands.

In a CNVS, the system reverses this logic:

- the existence of a state is not imposed by a governing authority, but is the result of an internal mathematical process;
- verification is entrusted to external verifiers, rigidly guided by a distribution model that uses restricted access to information as a method of collusion control;
- externally issued evidence must first pass strict internal consistency checks;
- state is not unitary;
- security emerges as a system property and increases as adoption grows.

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4. General Definitions

In this document, a System is defined as a structured set of objects and elements, regardless of type, possessing specific correlations and governed by preliminary assumptions that define its functions.

A System is considered C**losed** when no external intervention can alter its structure or produce outcomes that contradict its governing assumptions.

A System is defined as **Native Verification-Based** because the existence of all constituent elements and objects is admitted only after objective verification followed by a consistency check of the resulting attestations.

In other words:

> No element may come into existence unless it has first been verified.

A Native Verification System is defined as closed because no unauthorized external intervention can alter its structure or produce results that contradict its imposed assumptions.

For the purposes of the CNVS model, information (I_x) is defined as any finite ordered binary representation admitted by the System. Physical objects, measurements, attestations, certificates, signatures, and attributes are processed only through their Encoded Representation as elements of $\{0, 1\}^*$.

The System does not directly process physical Objects, but operates exclusively on their verifiable digital representation.

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5. Special Definitions

5.1 State and Structure

Let $E(t)$,

be the complete Set of all Structured Elements that compose the System at time t,
with $s_i \in E(t)$

Let $R(t)$,
be the Relations among the Structured Elements at time t,
with

$R(t) \subseteq E(t) \times E(t)$

Let C,
be the Deterministic Invariants imposed by the System, namely

$C = \{c_1, c_2, \ldots, c_m\}$

Let $V_{(\text{local or global})}$,

be the Internal Verification Function (local or global) for the Determination of Logical consistency before the declaration of the Existence of State.

A State will therefore be defined as:

$S(t) = [E(t), R(t), C, V_{Local}]$ **(1)**

Since Structured Elements are not atomic, but may be fragmented randomly without losing their Characteristics and Properties,
thus rendering the System Recursive, we will therefore have:

$E(t) = \{s_1, s_2, s_3,...., s_i\}$

with

$s_i = (id_i, D_i, At_{i,} Val_i, P_i)$ **(2)**

where,

$\forall\ id_i, \forall\ D_i \in (\{0, 1\}^*)^z$

$\forall\ At_i \in (\{0, 1\}^*)^x$

$\forall\ Val_i \in (\{0, 1\}^*)^y$

$\forall\ P_i \subseteq \mathbb{P}$

$\mathbb{P}$ is the finite or implementation-defined set of logical properties admitted by the system,

with $Z, X, Y \in \mathbb{N}$.

Therefore,

id_i = unique identifier,

D_i = data (e.g., 2 grams, 2 parts, or 2 mL),

At_i = Attributes (purity, internal state, timestamp, turbidity, etc.),

Val_i = Validations (audits, certificates, signatures, etc.) expressing Validated,

$\mathbb{P}$ = Properties,
{Valid, Transferable, Locked, Verified, Suspended, Redeemable,…}

with

$P_i = f(D_i, At_i, Val_i)$, or more precisely:

$P_i = V_{Local}(D_i, At_i, Val_i)$. **(3)**

Fundamental Property:

<u>decomposition does not change type</u>.

$s_{ij} \sim s_i$

Explicitly, we will have:

$s_{ij} = (id_{ij}, D_{ij}, At_{ij}, Val_i, P_{ij})$

with $\forall$ n, $\forall$ i, $\forall$ j $\in \mathbb{N}$

that is, the same and identical Structural Attributes,
Same nature,
Same schema.

Therefore:

$\exists \{s_{i1},\ldots,s_{ik}\} \subseteq E(t)$

i.e.:

$\{s_{i1}, s_{i2}, s_{i3}, \ldots., s_{ij}\} \subseteq E(t)$.

Hence

$\forall s_i \in E(t), \exists \{s_{ij}\} \subseteq E(t)$ **(4)**

such that:

structure (s_{ij}) = structure (s_i) **(5)**

Decomposition is <u>Finite</u> at every level.

Furthermore, during the Decomposition process, uniformity is not necessarily maintained, such that the obtained sub-elements preserve equal portions of the original Structured Element, but the Distribution of the Decomposition may be non-uniform.

Non-uniform Decomposition among Fragments of the same Level does not affect the Conservation of Information, but certainly affects the technical operation of its Reconstruction.

Formally:

$\forall s_i \in E(t), \forall n \in \mathbb{N}, \exists$ a decomposition of depth n **(6)**

or more precisely

$\forall s_i \in E(t), \forall n \in \mathbb{N}, \exists \{s_i^{(1)}, s_i^{(2)}, \ldots, s_i^{(n)}\}$ **(7)**

such that

level 0 of Origin

$s_i^{(0)} = \{s_1^{(0)}, s_2^{(0)}, \ldots, s_i^{(0)}\}$

level 1

$Sub(s_i^{(0)}) = \{s_{i1}^{(1)}, s_{i2}^{(1)}, \ldots, s_{ij}^{(1)}\}$

level 2

$Sub(s_{ij}^{(1)}) = \{s_{ij1}^{(2)}, s_{ij2}^{(2)}, \ldots, s_{ijl}^{(2)}\}$

with $\forall$ n, $\forall$ i, $\forall$ j, $\forall$ l $\in \mathbb{N}$

Therefore, for every n there exists a subsequent Decomposition level such that:

$s_i^{(n+1)} \in Sub(s_i^{(n)})$ **(8)**

and each $s_i^{(n)}$ preserves the defined Structure of the original Elements.

Explicitly:

$s_{ij} = (id_{ij}^{(1)}, D_{ij}^{(1)}, At_{ij}^{(1)}, Val_i^{(1)}, P_{ij}^{(1)})$.

Schematically, the sequence of Decomposition levels will be:

$s_i \rightarrow s_i^{(1)} \rightarrow s_i^{(2)} \rightarrow \ldots \rightarrow s_i^{(n)}$

where:

$s_1^{(0)} = s_1$ (initial fragment of level zero)

$s_1^{(1)} =$ fragments of $s_1^{(0)}$

$s_1^{(2)}$= fragments of $s_1^{(1)}$.

Therefore:

$\forall\ n \in \mathbb{N}, \exists\ s_i^{(n)} \in E(t)$, ideal model, **(9)**

$\forall \leq$ Depth(t), $\exists\ s_i^{(n)} \in E(t)$, operative model.

Within each sub-level, the portion of the Element from the preceding Level is not distributed uniformly.

In $s_{ij}^{(1)}$, the notation j as a subscript referring to subsequent Decomposition levels for the same fragment of Order i is replaced by the superscript (1), which assumes the same meaning as j.

In the formulae described from point (5) to point (6), both notations are reported.

N.B. In this Document the First Level of Structural Decomposition (Level 0) is called Nodal and the Structured Fragments si(0) of the Nodal Level are called Nodes.

5.2 Relations

Let

$R(t) \subseteq E(t) \times E(t)$ **(10)**

where,

$E(t) \times E(t)$

represents the Cartesian product of all possible pairs of the Structured Elements of Set E at the same time t, where R(t) is given by the

Directional Relations recognized by the System. If $E(t) = \{s_1, s_2\}$,

therefore

$E(t) \times E(t) = \{(s_1, s_1), (s_1, s_2), (s_2, s_1), (s_2, s_2)\}$.

The System does not necessarily provide for all possible Relations; only some may be permitted.
For example

let

$R(t) = \{(s_1, s_2), (s_2, s_2)\}$

The Directional Relation could only exist between
the pair s_1 and s_2
and between
the pair s_2 and s_2,
which would therefore only be in relation to itself.
The System therefore does not fully recognize all possible pairs of the Cartesian product E x E, but only some.
It is the Specific Implementation of the System, as well as the Structural Definition of its Constituent Elements, that decides on the Relations.
There are no Constraints Imposed by the System on the Relations, other than those of Logical Consistency to the Invariants and to the Global Structure of the System itself.

In the System

The Relation is always Directional (non-symmetric),
being the fundamental Order for Verification,
such that

$(s_j, s_i) \neq (s_i, s_j)$.

Put another way,

let

$\forall (s_i, s_j) \in R(t)$,

then

$$(s_i, s_j) \in R(t) \nRightarrow (s_j, s_i) \in R(t) \qquad \textbf{(11)}$$

All Relations allowed in the System must be
Logical,
Structural,
necessarily Directional (non-symmetric),
and non-Quantitative.
Relations are therefore Directional, and Connections are not weighted between Structured Elements.
The non-uniform Decomposition of Fragments does not change the Determination of the possible Relations between those Fragments at the same level of Decomposition.

The pair (E(t),R(t)) thus defines a directed graph over the structured elements of the system.

In this representation:

- each structured element $s_i \in E(t)$ corresponds to a vertex (node);
- each admissible relation $(s_i, s_j) \in R(t)$ corresponds to a directed edge;
- decomposition levels define a hierarchical structure in which nodes may generate descendant nodes through recursive fragmentation;
- nodal elements represent first-level vertices;
- fragmental elements represent vertices at deeper levels of decomposition.

The resulting Graph therefore captures both the logical relations among elements and the recursive structure of state decomposition.

5.3 Attributes (At_i)

Let

$$At_i = \{a_{i1}, a_{i2}, a_{i3}, \ldots, a_{ik}\} \qquad \textbf{(12)}$$

Attributes are observable data, not logical.

The Attribute is therefore an Objective Description of the Element.

Attributes may exhibit Internal Dependencies, but these cannot be modeled as Relationships within the System and are defined by the specific Implementation.

5.4 Property (P_i)

Properties are logical conditions that act on the single Structured Element, such that

$P_i = f(D_i, At_i, Val_i) = V_{Local}(D_i, At_i, Val_i)$ **(13)**

For example, a Property can be
valid, transferable, locked, verified, etc.
Properties are not data, but rather Evaluations that depend exclusively on the System.
Properties are therefore what $V_{Global\ or\ Local}$ uses or generates, such that

$\forall\ Pi \subseteq \mathbb{P}$ **(14)**

The Property is therefore a Logical Condition derived from the System that carries out the evaluation, therefore

$P_i = f(s_i)$, or rather

$P_i = V_{Local}(s_i)$ **(Local Validity)**

Property is therefore the result of a State Function.
Property is not limited to the definition of Validity, but also includes other logical conditions of the System that emerge from the development of its application.
For example, a si could be Valid, but also Blocked and Non-Transferable.
The formal definition of all possible Properties depends on the nature of s_i, which depends on the nature of S.
It is not possible to exhaust this topic in this document, so please refer to the Implementation for a complete description of the Parameter.
In this document, however, we can insist that the Property is not intrinsically

given, but must emerge from the outcome of the Verification Function applied to the State of the Structured Element.
The following paragraph will further clarify the meaning of the Function.

5.5 Validation (validated) and Validity (valid)

Let Val_i

The Certifications Issued by the External Verifier,
Validation therefore becomes an External Input to the System
and is part of the following Definition:

$s_i = (id_i, D_i, At_{i,} Val_i, P_i)$.

In short, it is the result of audits, signatures, and various certificates within the External Verification Process, and is performed by the External Verifier.
It contributes to the Evaluation, but does not exclusively determine it, because it is External to the System, even if guided and governed by it, and is not a Truth, but merely Evidence.
Validation is therefore an Input provided from Outside the System by the External Verifier.
Validation can result in a **"Validated" Condition**.

Valid $\neq$ 0

Valid is therefore a Structured Element that has received one or more Positive Validations, i.e., a Verification by an External Verifier that resulted in a Condition.
This does not mean it is Valid.
Validity is a Logical Property determined by the System, not the Validation condition.
Validity is therefore not an external input, but a result of the System.

$$\text{Valid}(S) \Leftrightarrow V_{Global}(S) = \text{TRUE} \quad \textbf{(15)}$$

In CNVS Systems, Validity is an Existence Constraint, such that

Valid ≠ Validated.

A State may be Validated but not satisfy the System's conditions (for example, it is duplicated, or there may be a Relationship error, an inconsistency with the Invariants, or it may generate a situation in which the Volumes or Quantities obtained exceed the Total calculated and expected by the System) and therefore not be Valid and therefore not become part of the System.
For a Structured Element that satisfies all the System's Conditions, therefore, its Verification Function will be:

$V_{Local}(s_i) = \text{TRUE}$.

In summary:

Validation (External Input) → System applies Function V_{Local}
Function V_{Local} → Validity (Property) → Systema applies Function V_{Global} (S).

5.6 Local and Global Properties

The Verification Function $V_{Global\ or\ Local}$ is applied not only to the State (S), but to all Fragments generated before they enter the State.
This produces a double Validity Property, Local and Global.

Validity $_{Local}$

Local Validity is a Logical Property referred to the Single Structured Element, such that

$V_{Local}(s_i(t))$

where

the Structured Element is individually verified at time t by the V_{Local} Function, which evaluates its Internal Structure, Attributes, Validations and Local Coherence.

Validity $_{Global}$

Global Validity is a logical property emerging from the consistency of the application of all the verification functions for each individual element and the resulting relationships between them and with respect to C.
Let

$V_{Global}(S(t))$

the Verification Function of the entire State S at time t, this Function checks the Relations R, the Invariants C, and the Consistency between the Structured Elements already verified by V_{Local}.
It is possible

$V_{Local}(s_i) = \text{TRUE}\ \forall_i$

but

$V_{Global}(S)$ = FALSE, because maybe the S violates the C.

Therefore

$V_{Local} \neq V_{Global}$

These are two Verification Functions applied to two different levels of the Structure, and therefore respond to different satisfaction criteria for successful completion of the inspection.
The relationship between the two, therefore,
is not

$V_{Global}(S) = \bigwedge_{si \in E(t)} V_{Local}(s_i)$,

but is

$$V_{Global}(S) \neq \bigwedge_{si \in E(t)} V_{Local}(s_i) \qquad \textbf{(16)}$$

that is, the Verification Function applied to the Whole State (global) is not equivalent to the Verification Function applied at the level of the Structured Element or its fragments (Local), but we will have

$V_{Global}(S) = f(E, R, C)$.

Therefore,
Local Validity refers to the individual correctness of the Structured Element,
Global Validity refers to the satisfaction of System Relations and Invariants, and cannot be reduced to Local Validity alone.
These are two different Functions applied to different domains: Local and Global.

5.7 V_{Global} and V_{Local} Verification Functions

Given the definition

$s_i = (id_i, D_i, At_i, Val_i, P_i)$

and

$\forall\, n \in \mathbb{N}, \exists\, s_i^{(n)} \in E(t)$,

The V_{Local} Verification Function will be applied to each si such that

$V_{Local}(s_i(t))$,

with

$V_{Local}(s_i) = \text{TRUE}\ \forall_i$.

The Function verifies the formal correctness of all parameters included in s_i, i.e., id_i, D_i, At_i, Val_i, generating the property P_i.
In detail, each of these Individual Parameters is verified for expressive correctness, local consistency, declaration plausibility, and internal logic.
For example, it may happen that the parameters D_i, At_i, and Val_i are all correct, but the system detects that the value of id_i (unique identifier) is impossible because it is expressed in an illegal formulation, or it may happen that the

parameters id_i, At_i, and Val_i are correct but the value of D_i is incorrect because it exceeds the maximum expected value for the decomposition level n considered (if the sum of all the values of D_i at decomposition level 3 has reached a value $D_i = 101$, and the sum of all the values of D_i from the previous decomposition level 2 was equal to 100, it is clear that there is a problem, because information cannot be generated from nothing, nor increased). In both cases, the value P_i that emerges from the application of $V_{Local}(s_i)$ will be FALSE, despite the correctness of all the other parameters.
It should be remembered that P_i not only includes a definition of Validity, but also includes other System Properties, for example Transferability.
Let

$V : S \rightarrow \{TRUE, FALSE\}$

S then corresponds to the Space of Possible States, such that

$S(t) = [E(t), R(t), C, V_{Local}]$,

then we have

$V_{Global}(S(t)) = TRUE$,

if and only if all the Conditions are satisfied.
This Function is to be understood as a deterministic function that evaluates whether a state satisfies the rules of the system,
i.e.

$\forall s_i \in E(t), V_{Local}(s_i) = TRUE$

R(t) is consistent
and

$\forall c_j \in C, c_j(S(t)) = TRUE$

therefore,

$$V_{Global}(S(t)) = TRUE \Leftrightarrow [\forall s_i \in E(t), V_{Local}(s_i) = TRUE] \wedge Consistent(R(t)) \wedge [\forall c_j \in C, c_j(S(t)) = TRUE] \quad \textbf{(17)}$$

where,

$V_{Local}(s_i)$, Local Verification of the Single Structured Element;

$R(t)$, Relations between Elements,
C, Global System Invariants.
So the V_{Global} Function uses
$E(t)$ to check the Elements,
$R(t)$ to check the Relations,
C to check the Invariants,
and the V_{Local} Function itself
to output a result.

The $V_{Global\ or\ Local}$ Function therefore does not create the Validation, but decides Local Validity if applied to the Structured Element, or Global Validity if applied to the State.
The $V_{Global\ or\ Local}$ Function is always deterministic, because the same State produces an identical result at time t.
Since the $V_{Global\ or\ Local}$ Function is defined for all States, we have

$$\forall\ S(t) \in S,\ V_{Global}(S(t)) \in \{TRUE, FALSE\}$$

Furthermore,

$$E(t) \subseteq \{ s \mid V_{Local}(s) = TRUE\} \qquad \textbf{(18)}$$

$$V_{Global}(S) \neq \bigwedge_{si \in E(t)} V_{Local}(s_i)$$

5.8 Protocol Parameters (ϵ, λ, I_0)

"Protocol Parameters" are defined as the set of critical coefficients (ϵ_G, ϵ_N, ϵ_C, λ, I_0) established by the System to regulate the Information Density and Fragmentation Tolerance Thresholds. These parameters are not independent variables, but structural constraints that determine the Validity or Rejection Status of a Node.

These parameters can be modified by the System or the Implementation to vary the expected thresholds and thus adjust the System Efficiency.
They will be discussed in the relevant chapters.

6. The State Transition Law

A state transition is defined as the evolution of the state from time t to time t + 1, such that

$$S(t) \rightarrow S(t + 1) \qquad \textbf{(19)}$$

Given the Transition Function T, we have that
$S(t+1) = T(S(t), I(t))$
where
$I(t)$ = external inputs (validations, events, etc.), or better

$$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1)) = TRUE\} \qquad \textbf{(20)}$$

The System therefore generates new S at time t+1, but there are no invalid intermediate states.
Only valid states that satisfy all the Conditions imposed for each constituent Structured Fragment become part of the System's State Set.
Furthermore, each S(t+1) must also pass the Global Validity check,
so the condition must also be true:
$V_{Global}(S(t+1)) = TRUE$.

That is, given that
$S(t+1) = S'$
it must satisfy that $V_{Global}(S') = TRUE$
with:

$$S' \subseteq T(S(t), I(t)) \qquad \textbf{(21)}$$

The Transition is therefore not free, but is constrained by verification.
In short, Transition States are constrained by the Verification Function, such that only Valid Structured Elements and Consistent States are admitted at each step, until they enter the Global System.

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7. Definition of Levels of Knowledge and Information

In a CNVS System, the accessible information is distributed along the decomposition imposed by the system itself, such that at each subsequent level, the size of the newly structured sub-element obtained is reduced, according to a recursive model that preserves the information structure, which is progressively rendered more incomplete.

The System does not generate Information Ix, but merely manages it after it has been entered. This premise is fundamental to understanding the User's role and the meaning of their request to the System:
Each Ix constitutes an External Request Input, such as a sales/purchase order, a request for objective verification, or a request for truth assessment, which queries the System as an Accreditation Device, which, after appropriate Verifications of all its component parts and fragments, determines whether this Input Request should be accepted and admitted or rejected.

The System will therefore receive a continuous flow of Information Ix, in the form of External Inputs, which will increase the IG (i.e., Global Information, see below) at time t.

By following the Information in this way, the Verifier's Knowledge can develop at multiple levels, with the same recursion.

It should be noted that within the system, information does not travel in a directly readable manner, but is encoded at the source so as to be cryptic.
The encoding is:

- systemic
- uniform
- structural.

Therefore, the fragmented information will not reach the verifier in a directly readable manner, but must first be decoded, because it will have inherited the encoding of the original information.
The system provides encoding as an additional security mechanism, but it is the implementation's responsibility to manage the model to be used.
In an ideal Model, decryption should be possible through the use of unique keys delivered exclusively to the final Verifier assigned to the specific execution Fragment.
In this way, even if an unauthorized Verifier were to access other Fragments, thus violating the exact system distribution map, they would be unable to read their contents, lacking the specific decryption keys for each individual Fragment.

<u>The information in the System is completely encrypted at every level.</u>
Put another way, the Verifier's access to all Fragments does not eliminate the encryption problem.

$\mathrm{Dec}(\tilde{I}_x(t)) \neq I_x(t)$ (see below, for the formalization of I_x)

for an unauthorized Local Verifier.

For simplicity's sake, the Model is formalized below by taking into account only the Final Encoding (Enc) notation relating to the Fragmental Information assigned to the Verifier, without reporting the notation for any level of

Information.
Given this, Knowledge K_x at time t can be understood as an accessible set of information, such that

$$K_x(t) \preceq I_x(t) \tag{22}$$

In detail, it is a subset of the information accessible to a subject, where

$I_x(t)$ = Existing Information

$K_x(t)$ = Accessible and known information,

for each domain x, at time t.

In this formulation, the Information Quantity Ix at time t derives from a Measurable Quantity of Structural and External Information Complexity to the System, and is not limited to the Fragmental content itself, nor to a specific Attribute, to the Unique Identifier, or to the data Di.

The Entire Representation is expressed within the System as a coded and ordered binary sequence {0, 1}, such that

$$I_x(t) \in (\{0, 1\}^*)^z \tag{23}$$

where:

- $I_x(t)$ is a finite, ordered bit representation;
- $\{0, 1\}^*$ is the set of all finite binary strings;
- x identifies the domain (global, nodal, fragmentary, local);
- $z, x \in \mathbb{N}$

Let $I_G(t)$ be the Global Information of the System at time t, such that

$$I_G(t) \cong (E(t), R(t), C) \tag{24}$$

So I_G fully preserves Structure, and is not just quantitative (although obviously I_G is ultimately still a sequence of bits).

The Information Coding $I_x(t)$ can be expressed as $\tilde{I}_x(t)$ at time t, such that

$$\tilde{I}_x(t) = Enc\,(I_x(t)). \tag{25}$$

Given K(t), the Knowledge achieved at time t,
the Global Knowledge K_G at time t is that relating to the Information I_G of the Entire System at time t,

let

$$I_G(t) = I(S(t)) \tag{26}$$

which corresponds to the Knowledge of the entire Structured Set E(t),
of all the Relations R(t)
and
of all the Constraints (C) imposed by the System,
we will then have

$$K_G(t) \preceq I_G(t). \tag{27}$$

Nodal Knowledge I_N at time t is that relating to the Information I_N relating to a generic First Node at time t, and therefore to all subsequent recursive Decompositions that depend on it up to all the fragments of the last level, such that:

$I_N(t) = I(N_i(t))$,

where
N_i constitutes the Node, so we have

$$K_N(t) \preceq I_N(t). \qquad \textbf{(28)}$$

The Level Knowledge $K_h^{(n)}$ at time t is that relating to the Information $I_h^{(n)}$ of all the Fragments that make up the same level at time t, but not the level that precedes it,

such that

$I_h^{(n)}(t) = I(s_{ij}^{(n)}(t))$

$$K_h^{(n)}(t) \preceq I_h^{(n)}(t) \qquad \textbf{(29)}$$

The Fragmental Knowledge K_F at time t is that relating to the Information If of the Single Fragment of the Last Level assigned by the System at time t, and corresponds to the Local Knowledge (Klocal) of the External Verifier,

such that

$$K_F = K_{Local}. \qquad \textbf{(30)}$$

Assume that Local Knowledge is Complete for that fragment, such that

$I_F(t) = I(s_{ij}^{(n)}(t))$,

so we will have

$$K_F(t) = I_F(t). \qquad \textbf{(31)}$$

The accessible Information Levels will have the following relationships:

$$I_F(t) \subsetneq I_h^{(n)}(t) \subsetneq I_N(t) \subsetneq I_G(t). \quad \textbf{(32)}$$

The K_{VF} Knowledge available to Verifier V_j at time t will be limited to its own fragment but total, so

$$K_{VF}(t) = K_F(t) = I_F(t) = I(s_{ij}^{(n)}). \quad \textbf{(33)}$$

On the fragmented Knowledge plane, however, there is no access even to the fragments that make up the Same Level.

Therefore, we will have

$$K_{VF}(t) \preceq I_N(t), \quad \textbf{(34)}$$

but

$$K_{VF}(t) \neq I_N(t)$$

and

$$K_{VF}(t) = K_F(t) \subsetneq K_N(t). \quad \textbf{(35)}$$

Furthermore,

$$K_N(t) \subsetneq K_G(t). \quad \textbf{(36)}$$

The Knowledge Relationships will therefore be

$$K_{VF}(t) \subsetneq K_h^{(n)}(t) \subsetneq K_N(t) \subsetneq K_G(t) \quad \textbf{(37)}$$

It should be noted that Fragment Assignment follows a random logic, such that a group of External Verifiers will not necessarily receive Fragments belonging to the same Level and the same Node, but External Verifiers may be assigned Fragments belonging to different Levels and Nodes.

That is,

$$Assign(f_{ij}^{(d)}) \rightarrow v_j \quad \textbf{(38)}$$

where $f_{ij}^{(d)}$ is a terminal fragment of Node N_i, but Verifier vj can receive fragments coming from different Nodes:

$$K_{VF}(t) = \{I_F(N_i, t), I_F(N_l, t), ...\}$$

with

$$i \neq l$$

This increases information dispersion, because the Verifier does not work within a recognizable "nodal class."

Furthermore, since the information is encoded, we will have

$$\tilde{I}_x(t) = Enc(I_x(t))$$

for all subsequent levels of Information up to

$$\tilde{I}_F(t) = Enc(I_F(t)). \quad \textbf{(39)}$$

The Verifier's Knowledge will be:

$$K_{VF}(t) = \tilde{I}_F(t)$$

with

$$\mathrm{Enc}(I_F(t)) = \tilde{I}_F(t) \preceq I_F(t) \tag{40}$$

Knowledge Level Table

Level	Meaning
K_F	fragment
$K_h^{(n)}$	level
K_N	node
K_G	system

Equivalence Table K, I

$K_F(t)$	$I_F(t)$
$K_h^{(n)}(t)$	$I_h^{(n)}(t)$
$K_N(t)$	$I_N(t)$
$K_G(t)$	$I_G(t)$

In short, Knowledge is not a simple quantitative reduction, but a Structural Reduction of the System's Representation, where it is possible to have a small but well-structured set of data, or by violating the System, to obtain much more, but poorly structured and poorly comprehensible data.

In both cases the available information is incomplete and fragmented to the point that reconstruction becomes difficult.

8. The Density of Information

Information Density ρ is a measure of the Dispersion of Information at the level in question and can be understood as the weight of the information at the level in question compared to the levels preceding it.

Information Density is represented at multiple levels.

The Local Information Density ρ_N is given by the measure of the contribution of Fragmental Information to the Node (or Nodal Density), and constitutes the portion of information per single Fragment compared to the Nodal Information.

It is a measure of the Dispersion of Nodal Information in Fragmental Information, such that

$$\rho_N(t) = I_F(t)/\ I_N(t) \qquad \textbf{(41)}$$

ρ_N measures how much the information of that fragment weighs compared to the information of its node (going up through all the levels of decomposition up to $s_i^{(0)}$) or how much the nodal information has been dispersed;

The Global Information Density ρ_G referred to the fragmental information is given by

$\rho_G(t) = I_F(t)/ I_G(t)$ **(42)**

ρ_G measures the weight of a fragment relative to the global information of the system.

Similarly, it is an index that measures the dispersion of global information in the terminal fragments.

The segmental information density ρ_{GN} is a measure of the importance of nodal information relative to global information and measures the weight of nodal information relative to global information, such that

$\rho_{GN}(t) = I_N(t)/ I_G(t)$. **(43)**

The Composite Information Density ρ_C is the composite measure obtained by taking into account both the effects of the Global Density and those of the Local Density, and constitutes a weighted geometric mean of two positive ratios,
such that

$\rho_C=(I_F/I_N)^\alpha * (I_F/I_G)^\beta$, **(44)**

where α and β are weighting coefficients that determine the relative contribution of the two information ratios, and that can be modified by the System or its Implementation to modify in the measurement of ρ_C the weight of the relative contributions of the Fragmental Information on the Nodal or Global Information, such that

$\alpha, \beta \in [0, 1]$ and $\alpha + \beta = 1$

and where

I_F / I_N measures how much the fragment weighs in relation to the Node.

I_F / I_G measures how much the fragment weighs in relation to System.

So: ρ_C

measures a Composite Information Density, which takes into account both the Fragmental exposure within the Node and the relative importance of the Fragment in the Global system.

System Consistency is guaranteed by compliance with the Protocol Parameters (see relevant section).

The System rejects any state transition in which the detected density exceeds the established threshold coefficients ($\rho > \epsilon \implies \text{Reject}(S)$) and violates the Conditions dictated by the Protocol Parameters.

---

9. Axioms

9.1 Axiom I

<u>In the System, only that which has been verified first and has obtained the Validity Property Condition exists.</u>

Anything that has not been verified first, or that, despite being verified, has not obtained the Validity Property Condition, simply does not exist for the System; that is, it is not permitted in the System, therefore we have

$$s_i(t) \in E(t) \implies V_{Local}(s_i(t)) = TRUE \qquad \textbf{(45)}$$

This has both Local and Global meaning.

On the Local level, we can say that a Structured Element si(t) belongs to the Set of Structured Elements of the System at time t if and only if it satisfies its Local Verification Function V_{Local}.

$$Valid\ (S(t)) \iff V_{Global}(S(t)) = TRUE \qquad \textbf{(46)}$$

With

$$S(t) = (E(t), R(t), C, V_{Local})$$

The Global State is valid if and only if the Verification Function applied to the State evaluates to TRUE.
In a Native Verification System, Existence is a Verification-Dependent condition.
A Structured Element becomes part of the System State if and only if it has satisfied the Verification Function.
Invalid Elements will not constitute the System State.

By also applying the Transition Function, we will have:

$$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1))=TRUE\} \qquad \textbf{(47)}$$

therefore,

$V_{Local}(s_i(t)) = FALSE \Rightarrow s_i(t) \notin E(t)$ **(48)**

in short,

$s_i(t) \in E(t) \implies V_{Local}(s_i(t)) = TRUE$

$V_{Local}(s_i(t)) = FALSE \Rightarrow s_i(t) \notin E(t)$

$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1))=TRUE\}$

$Valid\ (S(t)) \iff V_{Global}(S(t)) = TRUE$

9.2 Axiom II

<u>The System requires that the State be represented through successive Levels of Structured Decomposition.</u>

Let

$\forall\ s_i(t) \in E(t), \exists\ \{s_{i1}(t),\ldots,s_{ik}(t)\} \subseteq E(t)$

such that:

$s_i(t) \Rightarrow \{s_{i1}(t),\ldots,s_{ik}(t)\}.$

Let D be the Decomposition function

$\forall\ S(t), \exists$ a $D(S(t))$ Decomposition of depth n **(49)** such that:

$D^{(n)}(S(t)) = \{s_i^{(n)}(t)\}$

<u>the System imposes that the Decomposition Function D at time t increases the Structural Dispersion of the Global Information I_G as the latter increases.</u>

Given the Local Density ρ_N, Global Density ρ_G, Composite Density ρ_C,

The System adapts the Decomposition according to a Model that produces an increasing distribution of Information, with a progressive reduction of ρ_N, ρ_G, ρ_C, and therefore of the weight of Local Information over Global Information.

That is, if the nodal information grows, the System is forced to increase the depth of the decomposition for successive levels as the number of fragments increases, reducing the portion represented by each.

In summary:
The System imposes a Recursive Decomposition Constraint such that for each increase in both the Nodal Information In and the Global Information I_G, the relative share of Information accessible through the terminal Fragments is reduced such that,

$\rho_N(t) \leq \epsilon_N$ and $\rho_G(t) \leq \epsilon_G$

with

$\varepsilon_N, \varepsilon_G \in (0, 1]$

thus obtaining

$I_N(t2) > I_N(t1) \Rightarrow \rho_N(t2) < \rho_N(t1)$

and

$I_G(t2) > I_G(t1) \Rightarrow \rho_G(t2) < \rho_G(t1)$,

formally

$$(I_N(t2) > I_N(t1)) \lor (I_G(t2) > I_G(t1)) \Rightarrow Depth(t2) > Depth(t1) \qquad \textbf{(50)}$$

where ϵ_N, ϵ_G constitute two threshold coefficients, for ρ_N and for ρ_G, t1 and t2 represent two different time instants such that t1 < t2, and

$Depth \in \mathbb{N}$,

where Depth corresponds to the number of levels of the Recursive Decomposition applied by the System at time t.
The System imposes that as nodal information increases, the fragmentary information density with respect to the node decreases below a coefficient ϵ_N;
The System imposes that as global information increases, the fragmentary information density with respect to the Global System decreases below a coefficient ϵ_G.

In summary, the increasing Adaptive Decomposition therefore responds directly to the Composite Density, and indirectly to the Local and Global Densities, according to the following formulation:

$Depth(t)\uparrow \Rightarrow \rho_C(t)\downarrow$.

As the Composite Density increases, the Decomposition increases by successive levels, which then adapts to the increase in global Information by maintaining the Density levels or reducing them.

As the Composite Density increases, the Decomposition by successive levels increases, which therefore adapts to the increase in global information by maintaining or reducing the Density levels.

In fact, it is easy to demonstrate experimentally that the Densities ρ_N, ρ_G, and ρ_C progressively decrease as the System increases its Global and Nodal Information Level, regardless of any active interventions by the System itself.

This happens naturally, because Information Density is by definition a measure of a ratio, where the reference densities I_G and I_N are in the denominator:

In other words, if the size of the Information I_F at time t remains unchanged, while that of the Global Information I_G or Nodal Information I_N increases, each Fragment F containing the Information I_F, remaining of the same size, will have a smaller weight on the reference Information.

The effect will be a reduction in Density and therefore a Dispersion of Information, even without active interventions on the Fragments.
The size of each Terminal Fragment, which corresponds to the Fragmental Information I_F, is determined by the number of fragments imposed by the System, because the greater the Depth of fragmentation, the smaller the size of the Terminal Fragments will necessarily be and therefore the smaller the relative contribution of I_F Information.

This behavior is a problem, because instead of forcing the Decomposition to successive levels, it stabilizes it.

after applying the decomposition we will have instead

$\mathrm{Depth}(t)\uparrow \Rightarrow \rho_C(t+1) < \rho_C(t)$,

with $\rho_C(t) > \epsilon_C \Rightarrow \mathrm{Depth}(t+1) > \mathrm{Depth}(t)$.

The Condition "$\rho_C(t) > \epsilon_C$" then becomes the Internal input to the System which triggers further levels of Decomposition until the following effect is obtained:

$\mathrm{Depth}(t+1) > \mathrm{Depth}(t) \Rightarrow \rho_C(t+1) < \rho_C(t)$.

But the Condition "$\rho C(t) > \epsilon C$" is not easily verified experimentally because the growth of the I_F Information for a single Fragment does not follow that of the Information of the Whole System.

This happens because each Fragment remains isolated after decomposition, and while the system continues to expand into new nodes, the fragments already emitted remain stationary.
Even when the decomposition depth is forced, the effect is an uncontrolled drop in the Local Density level, such that further fragmentation is increasingly difficult.
In Chapter 10, new control coefficients will be introduced to address the problem.

We can therefore conclude that the densities ρ_N, ρ_G, and ρ_C constitute a measurement tool internal to the system, usable in implementation, rather than a set of control parameters.

Let's look at the number of Fragments in detail.

Let

$Fragments(t) \in \mathbb{N}$,

with $Fragments(N_i, t) = \gamma\,(I_N(t), \rho_c(t))$,

and $Fragments(t) = |\{s_i^{(n)}(t)\}|$

where

$Fragments(N_i, t)$ corresponds to the number of Terminal Fragments generated by the Adaptive Decomposition process at time t in Node N_i,
$Fragments(t)$ corresponds to the number of global fragments at time t,

γ it is a monotonic function increasing with respect to information complexity,

The Decomposition applied to the Node must depend not only on the internal Information Breadth, but also on its relative Information Weight with respect to the Global System.
So the level of Decomposition is a direct function of the Composite Density measure, which depends on the Local and Global Density Values, such that

$$\rho_C(N_i, t) \leq \epsilon_C \qquad \textbf{(51)}$$

where

$\varepsilon_C \in (0, 1]$

The System must force the Depth of Information Decomposition to maintain Dispersion and reduce Information Density.

This happens when the Composite Density ρc increases, because it means that the Terminal Fragment weighs too much compared to the node/global. Therefore, the System is forced to increase the Decomposition Depth to reduce that density and preserve the Structure.

Let

$$\text{Depth}(t) = f(\rho_C(t)) = f(I_N(t), I_G(t)) \tag{52}$$

with f being a composite Density Function, we will have

$$K_{VF}(t+1) < K_{VF}(t) \tag{53}$$

and

$$K_F(t) \subsetneq K_G(t) \tag{54}$$

The Decomposition grows monotonically with respect to both I_N and I_G.

Since ∀ S(t), ∃ a decomposition D(S(t)) of depth n

the Decomposition level requires a number of External Verifiers, such that

$\rho_c \leq \epsilon_c \Rightarrow D(S(t))$ *requires* $|V_{ext}^{required}(t)|$

where

- $|V_{ext}^{required}(t)|$, set of Verifiers required at time t

- ρ_c , Composite Density

- ϵ_c , the fragmentation coefficient

therefore,

$$|V_{ext}^{required}(t)| \geq |Fragments(t)| \qquad \textbf{(55)}$$

that is, the System requires a sufficient number of Verifiers to sustain the imposed level of Decomposition D.

This is because Decomposition is an Epistemic Property of the System, not an accidental dependence on the Number of Verifiers available at time t.
In other words, the System does not depend on the number of Verifiers, but manages it and requires it as a Preliminary Condition.

Let's clarify that within the System, the Verifier is not the User.

9.3 Axiom III

<u>The Validity of the System cannot be reduced to the Validity of the Individual Structured Elements that constitute it.</u>

$$Valid(S(t)) \Longleftrightarrow V_{Global}(S(t)) = TRUE$$

$S(t) = (E(t), R(t), C, V_{Local})$ and $V_{Global}(S(t)) = f(E(t), R(t), C, V_{Local})$,

therefore

$$V_{Global}(S(t)) \neq \bigwedge_{si \in E(t)} V_{Local}(s_i(t)). \quad \textbf{(56)}$$

Let's look at it in more detail, because at first glance it might seem like a Law deduced from the previous Axioms, but in reality it has its own Dignity that requires the definition of an independent Invariant.

Given that

$\forall s_i \in E(t), V_{Local}(s_i) = TRUE$

it may still occur that

$V_{Global}(S(t)) = FALSE$

This happens because the Validity of the State depends not only on the Verification itself, but also on R and C.

Therefore,
$V_{Global}(S(t)) = TRUE$

if and only if

$[\forall s_i \in E(t), V_{Local}(s_i) = TRUE] \land Consistent(R(t)) \land [\forall c_j \in C,$

$c_j(S(t)) = TRUE]$ **(57)**

This Axiom is extremely important for ensuring Global Structural Coherence, because without this Invariant Specification, the System would break down simply because successive Local Verification Aggregation Conditions would be sufficient to obtain the Valid State, without imposing a Global Coherence Check at higher levels.

In other words, the System is structural and does not exhibit aggregative properties for different Verification Levels.
Therefore, Validity is an emergent Property, not a simple sum.
R and C are essential for the State Definition.
In summary:

Global Validity cannot be reduced to Local Validity. **(58)**

Verification of the System's State depends not only on the Individual Structured Elements, but also on the Consistency of Relations and compliance with Invariant Constraints.
In conclusion, Axiom III establishes that in a CNVS System, Global Validity must also depend on Relations and Invariants, not just on the Structured Elements.

Therefore:

$$V_{Global}(S(t)) \neq \bigwedge_i V_{Local}(si(t)), \quad \textbf{(59)}$$

this is because the Verification of a State, on which its Validity depends, is not only the Derivation of the Aggregation of the Verifications of the Individual Elements that constitute it.

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10. Information Restriction Principle

Introduction

Since the decomposition depth does not respond directly to the information, but responds to the resulting information density, the relationship is:

$\rho_C(t) \le \epsilon_C$,

therefore, as Recursive Depth, Information Dispersion, and Adaptive Fragmentation increase, the relative share of locally accessible Information progressively approaches zero with respect to the Node and the Global System.
In Chapter 7, the problem of the difficulty of managing Fragmentation if controlled directly through Composite Density was raised.
The proposed solution is to control the Decomposition Depth under the reciprocal of ρC, assuming Uniform Nodes and Fragments, such that:

$1/ (\rho_C(t) * N(t)^{\beta})$

Furthermore, considering that the growth of I_G must also be able to take control of it, we have:

$\log_2 (1+(I_G))$.

The two relations can be operationally extended by introducing the Fragmentation Intensity Coefficient λ and the reference Information Unit I_0, resulting in the following Fragmentation Function defined:

$$\text{Fragments} = \lceil \lambda \cdot [1/ (\rho_C(t) \cdot (N(t)^{\beta)})] \cdot \log_2(1 + (I_G(t) + I_N(t))/ I_0)] \rceil \qquad \textbf{(60)}$$

where

$\lambda \in \mathbb{R}_{>0}$ e $I_0 \in \mathbb{R}_{>0}$ e $N \in \mathbb{N}$

such that

λ, constitutes a Fragmentation Intensity Coefficient;

ρ_C, the Composite Density understood as a weighted geometric mean;

N, represents the total number of active Nodes already present in the System, i.e., the cardinality of E(t),

with $E(t) = E_{int}(t) \cup E_{term}(t)$

$E_{int}(t)$, the internal and structural Nodes up to the terminal level;

$E_{term}(t)$, the terminal Fragments only,

$N(t) = |E_{int}(t)|$ N, i.e., at time t

β, a Protocol Parameter;

I_0, a reference information threshold, indicating the interval in bits for each increase of +1 fragment, i.e., a positive reference unit of information that determines the scale at which increases in global information trigger further fragmentation. Its value is defined by the implementation;

$I_G(t)$, $I_N(t)$ already defined previously,

all at time t.

With the formula in (60) the imposition of growth of the number of Fragments does not depend only on the number of initial Nodes, therefore on their variation, but also on the growth of the Global Information I_G, and evidently on the Nodal Information I_N independent of how it is distributed in the successive nodes.

The System or Implementation may provide for a variation of the various coefficients, including λ, to manage the efficiency of fragmentation.

The tables at the end of this document show concrete examples.

Definition

Let:

$K_{VF}(v_j)$

the Knowledge accessible to the External Verifier vj at time t
and lets

$I_F(t)$, $I_N(t)$, $I_G(t)$

Fragmental Information, Nodal Information, and Global Information, respectively,

Recalling that $I_G(t) = (E(t), R(t), C)$
and the System follows the Imposition to the Increasing Depth of Decomposition n, the formulation

$$\text{Fragments }(t) = F_x(t) = \lceil \lambda \cdot [1/ (\epsilon_C \cdot (N(t)^{\beta)})] \cdot \log_2(1 + (I_G(t) + I_N(t))/ I_0)] \rceil$$

where

$F_x(t) \in \mathbb{N} \geq 1$

with

$\lambda \uparrow \Rightarrow F_x \uparrow$
$I_G \uparrow, I_N \uparrow \Rightarrow F_x \uparrow$

$I_0 \downarrow \Rightarrow F_x \uparrow$
$N(t) \uparrow \Rightarrow F_x \downarrow$

and therefore $K_{VF}(t) \downarrow$

It follows that:
<u>The System, through its invariants, imposes a Distribution of Information such that the Local Knowledge K_{VF} accessible to each External Verifier vj is structurally insufficient for the reconstruction of the Nodal Information and the Global Information.</u>
The Knowledge Restriction becomes the Structural Condition of the entire System.
Formally:

$$K_{VF}(t) \nRightarrow I_N(t) \quad \textbf{(61)}$$

and

$$K_{VF}(t) \nRightarrow I_G(t) \quad \textbf{(62)}$$

Therefore, Local Knowledge K_F does not contain, allow, or imply the information reconstruction of the Node or the global System.
In Detail, given the Adaptive Decomposition

$D^{(n)}(S(t)) = \{s_i^{(n)}(t)\}$

and the progressive reduction of densities

ρ_N ,ρ_G, ρ_C,

with

$\rho_N(t) = I_F(t)/ I_N(t)$

$\rho_G(t) = I_F(t)/ I_G(t)$

$\rho_C=(I_F/I_N)^{\alpha} * (I_F/I_G)^{\beta}$

$\alpha, \beta \in [0, 1]$

with $\alpha + \beta = 1$, established that $t2 > t1$,

and after the Application of the adaptive fragmentation function

$\forall\, t2 > t1, I_N(t2) > I_N(t1) \Rightarrow \rho_N(t2) < \rho_N(t1)$

and

$\forall\, t2 > t1, I_G(t2) > I_G(t1) \Rightarrow \rho_G(t2) < \rho_G(t1)$

from the application of the imposed Axioms it emerges

$I_F(t) = I_h^{(n)}(t) / F_x(t)$

where

$I_h^{(n)}(t)$

constituting the leveling information Ih(n) of all the Fragments that make up the same level at time t, but not the level preceding it, and $K_h^{(n)}$ its leveling knowledge, we have that

$I_h^{(n)}(t) = I(s_{ij}^{(n)}(t))$ e $K_h^{(n)}(t) \preceq I_h^{(n)}(t)$,

with

$I_F(t) = \bar{I}_F(t)$

which constitutes the average size of fragmented information, since we have assumed that they are not uniform among themselves,
the relationship therefore becomes

$(\rho_C, \lambda, I_0) \Rightarrow F_x(t) \;\Rightarrow\; I_F(t) \Rightarrow K_{VF}(t)$

and since

$K_{VF}(t) \leq I_F(t),$

it turns out that:

$$\lim_{Fx(t)\to\infty} K_{VF}(t)=0 \qquad \textbf{(63)}$$

and consequently,

$$\lim_{Fx(t)\to\infty} \rho_N(t)=0 \qquad \textbf{(64)}$$

Information limitation ultimately constitutes a form of structural insufficiency of Local Knowledge with respect to the Global complexity of the System, such that the complete informational reconstruction of the Fragment, of all the levels preceding it, and of the Global one, remains an emergent property exclusive to the System.
<u>The Principle of Information Restriction therefore becomes an Epistemic Property deriving from the Application of the Axioms.</u>
Formally, since

$D^{(n)}(S(t)) = f(F_x(t)) = f(\rho_C(t), \lambda, I_0, N(t))$

we conclude that

$$K_{VF}(t+1) < K_{VF}(t) \tag{65}$$

The lower the allowable Composite Density, the greater the depth, fragmentation, and information dispersion must be: that is, the System increases the Decomposition to reduce the accessible Information Density, based on the set Protocol Parameters, over which it has full control.
This implies that the reduction of Knowledge can be "regulated" within the System as a Condition whose Threshold level depends on the Coefficient ϵC and how this affects the imposed Value of ρ_C and overall on all the Protocol Parameters, including α, β, and I_0.

Let $H(I_N)$
be the Information-theoretic Entropy of the nodal information, measured in bits.

The Knowledge Restriction Principle requires that the Conditional Entropy of the nodal information, given the fragmented knowledge K_{VF} available to a single verifier, remain close to the original entropy:

$$H(I_N \mid K_{VF}) \approx H(I_N) \tag{66}$$

Ideally, the information obtained about the complete Nodal State is limited, and the Verifier's knowledge of the Fragment should not significantly affect the Uncertainty across the entire Node, because the verifier should learn virtually nothing about its overall content through fine-grained fragmentation.
This can be expressed using Mutual Information:

$$I(K_{VF}; I_N) \approx 0,$$

that is, each Verifier knows only a limited and increasingly smaller portion of the overall Content, so we will have

$$H(I_N | K_{VF}) = H(I_N) - I(I_N; K_{VF})$$

$H(I_N)$, total entropy of the node;

$H(I_N | K_{VF})$, residual entropy after observing the fragment.

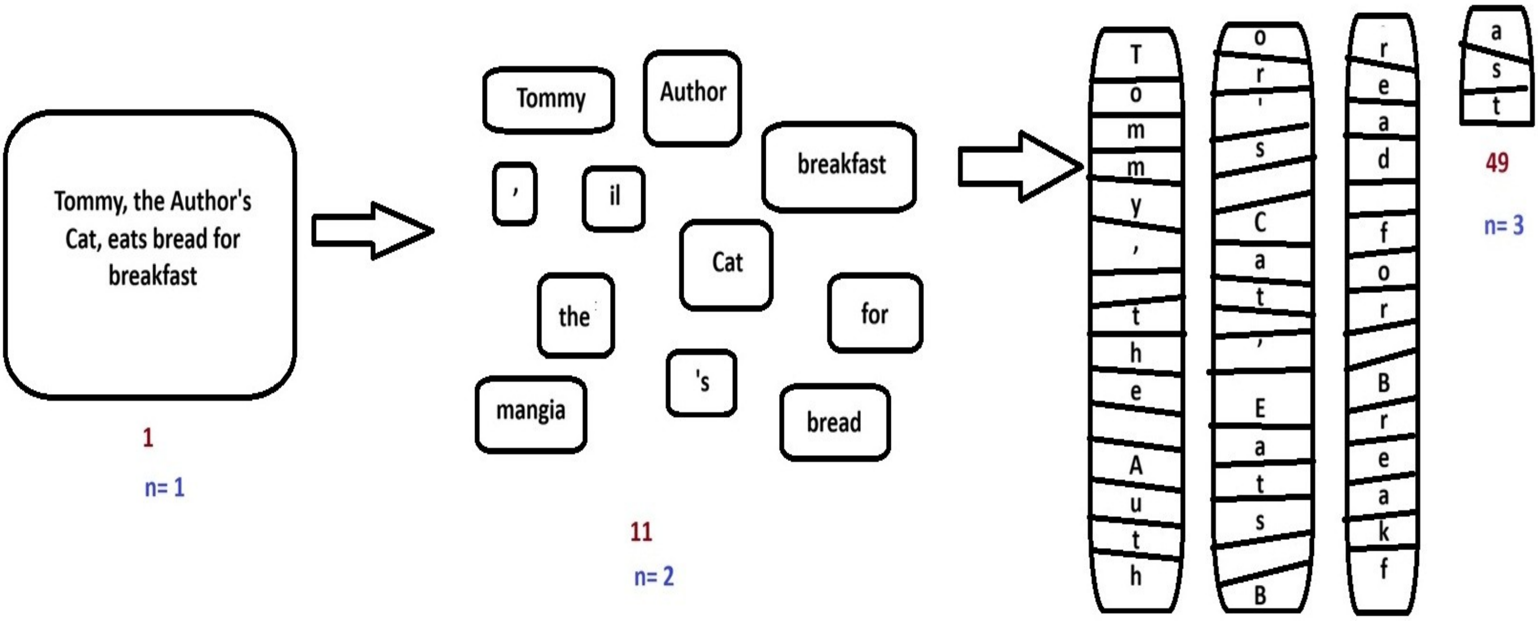
This means that access to a single local fragment does not significantly reduce the uncertainty about the complete nodal state, because the system is designed to minimize the mutual information between the verifier's local knowledge and the complete nodal state.

---

11. Discussion on Shannon Entropy

Formalizing the behavior of Shannon Entropy in CNVS Systems is a central point of the theoretical development.
The System Architecture is based on the premise that the progressive reduction of the information content of the terminal I_F Fragments, according to the Knowledge Restriction Principle, due to the information granulation of the original I_G Information, makes its reconstruction increasingly difficult for the External Agent.
In an ideal model, by repeating the fragmentation ad infinitum, the information content of I_F becomes completely devoid of any meaning for the External Agent:
the relationships between the I_F Fragments thus tend to progressively reduce to randomness, losing all logical connection.

An example might clarify the conceptual meaning.
Let's look at the Box below.



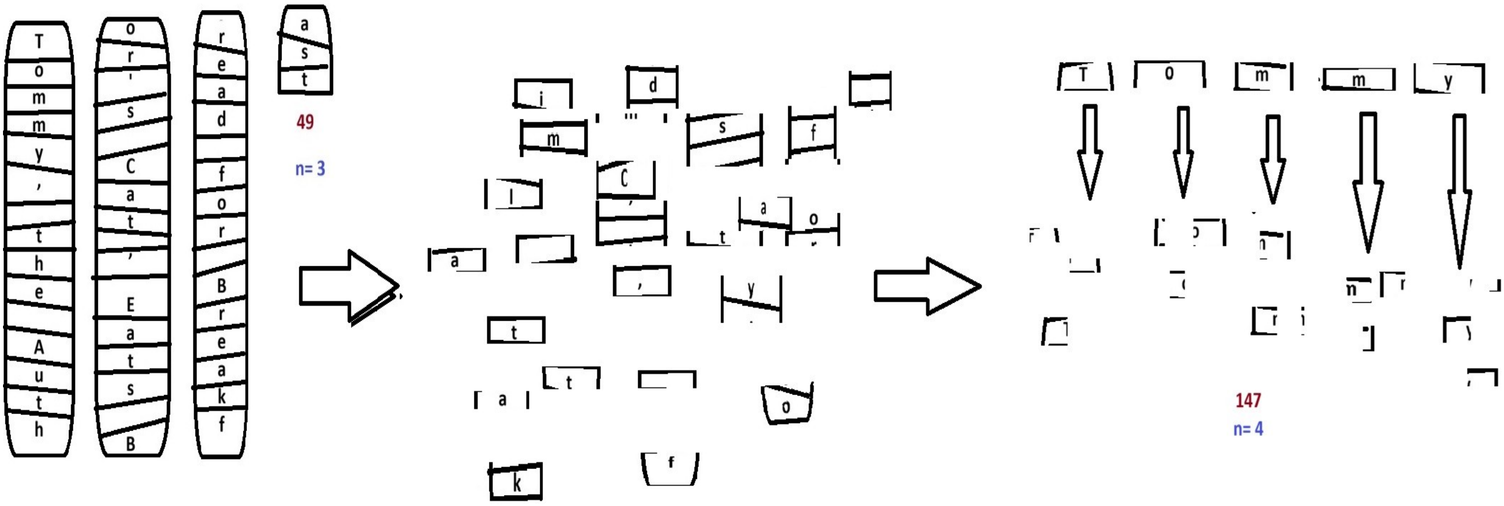
It seems clear that the sentence "Tommy, the Author's Cat, eats bread for breakfast" makes logical sense to the External Agent: the Information begins with a series of intuitive logical relationships that anticipate and facilitate its reconstruction.

The situation on the left side of the Box appears quite clear: one enormous fragment at Level 1, conceptually strong.

In the central section of the Box, we already see something interesting: following the first informational granulation, the sentence's content is decomposed, producing 9 fragments at level 2.

On the right side of the Box, we see something even more interesting: following the second informational granulation, the sentence's content becomes highly decomposed, producing 49 fragments at level 3. At this level, the message is no longer clear, but the semantic content of each individual part still has meaning for the reader, even if it lacks any intuitive logical relationship: they appear to be just scattered letters (T, o, c, l, z, etc.).

Let's continue our reasoning and see what happens at level 4, in the new Box below:



The result is a box containing a set of objects devoid of logical meaning, which have even lost their semantic meaning.

The knowledge of an External Agent with level 4 fragments will be even more limited, because the absence of any logical and relational constraints between them makes reconstruction extremely difficult.
If we were to continue the granulation process virtually infinitely, we would obtain a fragmentation of the I_G information such that the I_F would become so destructured as to make reconstruction increasingly technically difficult, to the point of becoming virtually impossible in extreme cases.

This example shows how the CNVS system, using the decomposition of the source I_G information, produces the effect of reducing the information content of each individual fragment obtained I_F, such that the I_F content becomes a function of the number of fragments.

Approximating, we could say:

$K_F = I_F \approx I_G / n$

such that

$K_F \rightarrow 0$ as $n \rightarrow \infty$

Formally,
the information content of I_G can be interpreted in terms of progressive decomposition.
In an ideal CNVS system, starting from a certain granulation threshold (Limit Threshold), the semantic, logical, and relational meaning of its content is completely lost, and its exact recomposition follows only probabilistic logic.

Assuming that the reconstruction problem consists in identifying the correct ordering of n distinguishable fragments among all equiprobable permutations, the probability of a successful random reconstruction is:

$p = 1/ n!$

If I_G Decomposition is applied in cycles such that its content is progressively fragmented twice at each cycle, the situation will be as follows:
after the first granulation cycle, we will have
I_G divided into two fragments, each containing 1/2 of the original information.

Formally,

$I_F(t) = I_G(t)/ 2$ with $p = 1/ (2*1) = 0.5$

If we apply 3 cycles, we will have:
$I_{F,3}(t) = I_G(t)/ 2^3$ with $p = 1/ (6*5*4*3*2*1) = 0.00138888$

If we repeat the cycle 20 times, we will have:
$I_{F,20}(t) = I_G(t)/ 2^{20}$ with $p = 1/ (2^{20})!$,

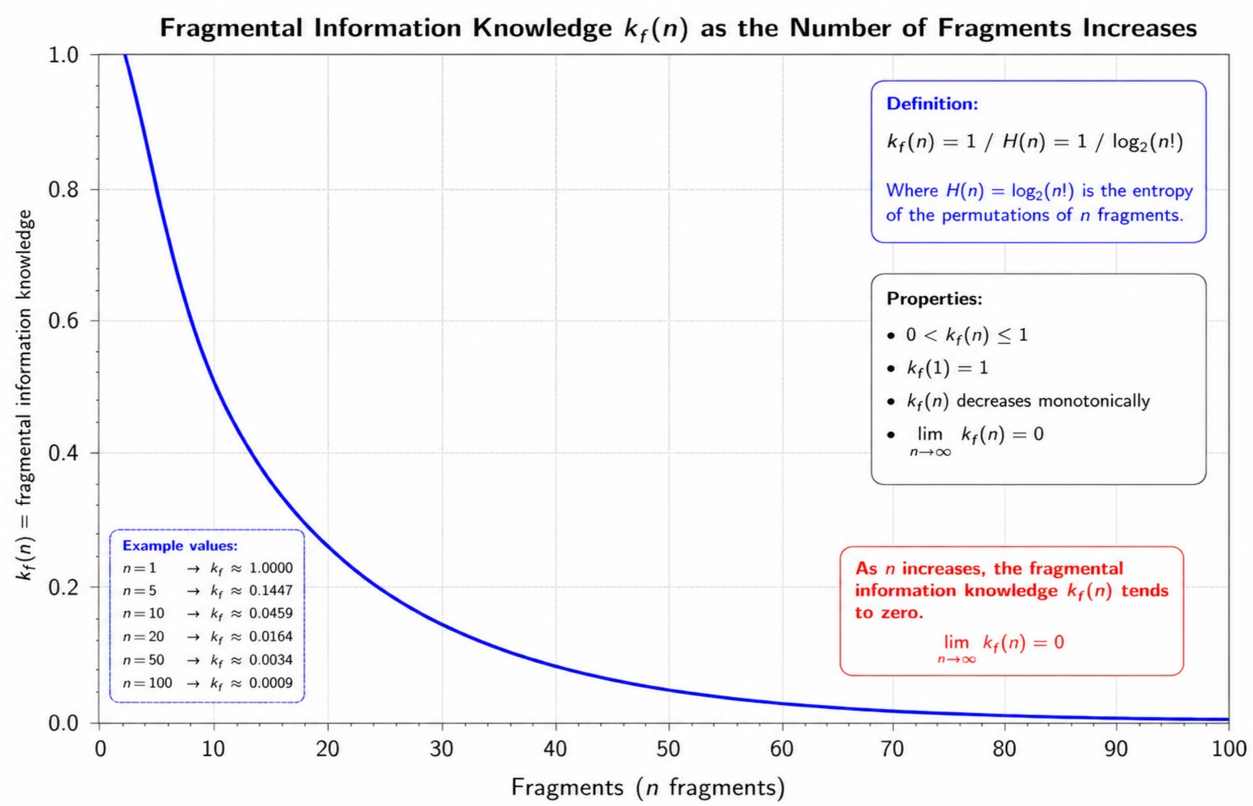
generically,

$I_{F,m}(t) = I_G(t)/ 2^m$ and $p_m = 1/ (2^m)!$

with $m \in \mathbb{N}$,

m is the number of decomposition cycles.

It is clear that the probability of finding the right combination by relying solely on chance becomes so low that the effort is completely useless.
Graphically, we can represent the relationship between the Knowledge of Fragmental Information $K_F(n)$ and the number of Fragments (n), with $n \rightarrow \infty$



In this context, Shannon entropy measures the uncertainty associated with the reconstruction problem, namely the uncertainty of identifying the correct configuration among all equiprobable permutations of the fragments.

We calculate Shannon Entropy in a model in which we have 20 fragments arranged in a row and each recognizes a single successful combination among all equally likely combinations.
Let $p_i = 1/ 20!$,

let Shannon Entropy be

$H(X) = -K \Sigma_{i=1}{}^{N} p_i \log_2 p_i$,

N= 20!, K= 1

replacing, $H(X) = - \Sigma_{i=1}{}^{20!} 1/ 20! \log_2 (1/ 20!)$,

and since the term is the same for all combinations we will have

$H(X) = -20! * 1/ 20! \log_2 (1/ 20!)$,

$H(X) = - \log_2 (1/ 20!)$

$H(X) = \log_2 (20!)$

$H(X) = 61.08$ bit.

By composing the 20 fragments one at a time we will have

20*19*18*⋯*2*1

the correct probabilities at each step become

1/ 20 * 1/ 19 * 1/ 18 * 1/ 17 *…. * 1/ 2 * 1/ 1.

then,

$p = 1/20!$

the total information will be

$I = -\log_2(P)$

$I = \log_2(20) + \log_2(19) + \log_2(18) + \cdots + \log_2(2) + \log_2(1)$

$I = \log_2(20!)$

$I \approx 61.08$ bit,

In conclusion,
P(correct combination by chance) = 1/ 20!
20! = 2,432,902,008,176,640,000.
Therefore, by randomly choosing an arrangement of the 20 fragments, the probability of immediately obtaining the correct one is approximately:
$4.11 * 10^{-19}$
that is, approximately $4.11 * 10^{-17}$% probability of guessing the correct combination on the first try.

Examining the data from the Binary Information relating to the calculated Uncertainty, we will have that to uniquely identify the correct arrangement among all 20 possibilities, approximately 61 binary bits of information are required [1, 0].
61.08 bits correspond to $2^{61.08}$ possibilities.
And in fact, $2^{61.08} \approx 20!$ corresponds exactly to the number of permutations of 20 objects.

Let's try to extend the reasoning by gradually increasing the Decomposition up to the limit $+\infty$ and see how the Entropy behaves.
The number of possible combinations/permutations is:
N(n) = n!
If all are equally probable:
p_i= 1/ n!,
substituting into the Shannon Entropy we have

$$H(X) = - \Sigma_{i=1}^{N} p_i \log_2 p_i$$

$$H(X) = - \Sigma_{i=1}^{n!} 1/ n! \, \mathrm{Log}_2 (1/ n!)$$

therefore

$H(X) = - n!\ 1/ n!\ Log_2 (1/ n!)$, and simplifying

$H(X) = - Log_2 (1/ n!)$,

$H(X) = Log_2 (n!)$.

Let's calculate the limit

$\lim H(X)_{n\to\infty} = \lim_{n\to\infty} Log_2 (n!)$

with $n \to \infty$, then $\log_2(n!) \to \infty$

$\lim H(X)_{n\to\infty} = +\infty$ **(67)**

Since n! becomes a very large number, we perform the calculations using Stirling's approximate formula, which was created specifically to handle these special cases.
The formula is $n! \approx \sqrt{(2\pi n)} * (n/ e)^n$

so we will have

$H(n) \approx \log_2 [\sqrt{(2\pi n)} * (n/ e)^n]$

$H(n) \approx n \log_2 n - n \log_2 e + ½ \log_2 (2\pi n)$

The dominant term is:

$n \log_2 n$

$H(n) \to +\infty$

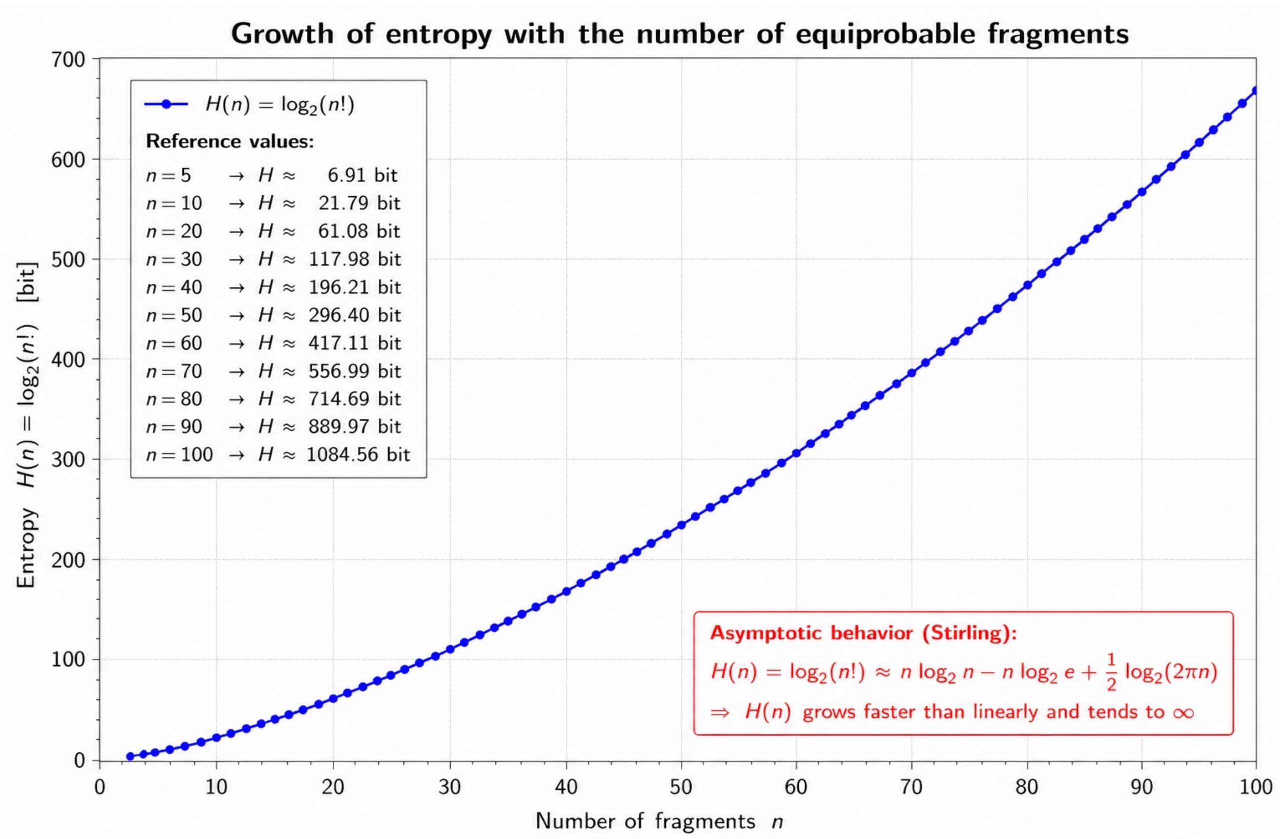
the probability instead will be

p = 1/ n!.

Therefore

$\lim_{n\to\infty} P_n = \lim_{n\to\infty} 1/ n! = 0$

Therefore, as the number of fragments grows without limit, the probability of randomly guessing the correct combination approaches zero, while entropy approaches infinity.
Graphically,



Therefore, in an ideal CNVS model, increasing fragmentation causes the fragmental information $I_F(n)$ to decrease, while the entropy associated with the reconstruction problem increases. However, the system cannot pursue infinite fragmentation as an absolute objective, because below a certain operational threshold the fragment would no longer preserve the minimum information required for verification. The implementation must therefore identify an optimal threshold n^*, as a function of the source information I_G, at which reconstruction uncertainty is maximized while fragmental information remains operationally usable.

Concluding,

$I_F(n) \geq I_{F\_min}$

with I_{F_min} denotes the minimum amount of fragmental information required for the fragment to remain operationally usable for verification

and with the threshold value thus formalized:

$n^* = \max\{n \mid I_F(n) \geq I_{F\_min}\}$.

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12. State Rejection

The System is designed to resist External Hacking and Authoritarian Manipulation attempts, but there may be Situations, discussed in a separate document, that cause integrity failure.
These Situations can occur when certain Conditions that the System deems essential for its very survival cease to exist.

These Conditions at time t include:

- Density can no longer be maintained;
- Verifiers failure;

- Depth does not increase sufficiently;
- ρC exceeds the threshold ϵ_C;
- ρN exceeds the threshold ϵ_N;
- ρG exceeds the threshold ϵ_G.

The System also is rejected when the Internal Verification process is determined by the External Verification process in such a way that it becomes:

- predictable;
- manipulable;
- subject to collusion.

In all these cases, we will have

$V_{Global}(S)$ ≠ deterministic and therefore

the Existence of S ≠ inherent to Verification.

In all these cases, the System loses the right to be considered a CNVS and becomes a Traditional System, because the Integrity of the Verification Function is not a functionality, but a definitional requirement, such that the System is not only violated, but becomes a different type of System.
The Integrity of Verification is not a functional requirement, but an necessary precondition for membership.

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13. Limitations of the Theoretical Model

External Verifiers constitute a Membership Class, which is not necessarily "human." Technically, External Verification can be entrusted to the use of hardware devices supervised by human professionals, or it can be completely automated.
This Document does not go into detail about the methods used to perform External Verification, but it assumes that it is performed with Objective Judgment, structuring a System that reduces the possibility of involuntary and intentional errors through Fragmentation Control and the Reconstruction of the Global Asset with its progressive implementation.
This is even more true considering that every External Verification is permanently archived within the System, which preserves the Historical Memory of each Individual Verification for each Individual Structured Element.
The ability to track the Unique Identifier of each External Verification, as well as all individual steps within the System, makes a Violation theoretically impossible without traceability.

External Verifiers can therefore be:

- Human auditors
- Automatic validators
- Sensors
- Vaults
- Software agents
- Distributed modules.

Specifically, the sustainability of decomposition growth could become a long-term problem.
If the value of Fragments(t) were to grow too rapidly, then the computational cost would become difficult to offset.

At the same time, there could be issues with synchronization, latency, and relationship management.

Systemic Complexity can be well summarized by reading the following comparison table:

More fragmentation	Less knowledge
less knowledge	More emergent security
More fragmentation	More systemic complexity

The model shows a natural tradeoff, which could constitute a serious limitation for its application.

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14. Further Developments

In a subsequent implementation development, an optional extension could be envisaged.
In the Extended Model, the introduction of a Functional Micro-Decomposition would significantly increase restriction of knowledge, because it would fragment the I_F Information even more extremely.

Let's assume we further fragment the I_F following the formulation

$I_F(t) = (q_1,q_2,\dots,q_m)$.

Each External Auditor could access only

$K_{VF}(t) = Enc(q_m)$,

with $K_{VF}(t) \nRightarrow I_F(t)$

since the Verifier can no longer directly access $Enc(I_F(t))$.

This is because the terminal Fragment would not be assigned as a Complete Structured Unit with I_F content, but would be Functionally Destructured into its constituent Components, such that

$I_F=(q_1,q_2,\ldots,q_m)$

where q_1, q_2,…, q_m constitute the elementary functional components of the I_F Information (for example, q_1 = height, q_2 = weight, q_3 = purity, q_4 = volume, etc.)
From this perspective, qm is no longer an autonomous Structured Fragment with a "complete" meaning, but an elementary functional verification component.
In this advanced Model, the Verifier does not know the node, does not know the level, does not know the fragment, does not know the complete operational meaning even of what it is measuring, but only knows a Specific Elementary Verification Function, as the reconstruction of the Fragment itself that carries the I_F Information becomes an emergent systemic property.
Thus, the reconstruction of the Structured Fragment occurs only at the System level.
Formally

$I_F(t) \notin K_{VF}(t)$

but

$q_m \in K_{VF}(t)$

and

$I_F = \Phi(q_1, q_2, \ldots, q_m)$

where

Φ

is the reconstruction function managed by the system.

New Information Levels Table

Level	Rebuildable from
q_m	Local Verifier
I_F	System
I_N	Higher Layer
I_G	Global System

In this model, the verifier is not even able to fully understand the meaning of the component it is measuring.

This can have interesting implications, as it significantly increases the security of the system, without violating the CNVS theory, but by fully integrating it into its architecture.

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15. The Emergent Security Theorem

Introduction

In this document, the Security of a System is defined as the Property of the System to resist attacks by Colluding External Agents attempting to violate it.
In this document, the term "Emergent" is used to indicate a Property of the System that emerges as a consequence of its intrinsic structure and is strengthened by its progressive adoption.
In this sense, the Emergent Security Theorem defines the Property of the System to resist attack attempts by Colluding External Agents that share the I_F Information assigned by the System with each other, in order to violate it.
The Theorem proposes an emergent Property such that the intention of the Colluding Agents clashes with the growth of the System's Resistance to violation with its progressive adoption and with the increase in the I_G content and therefore its complexity.
To understand the Theorem, we must use the tools that mathematics offers us.
Let's imagine a situation in which a number n_{bad_agent} of Colluding External Agents wants to breach the System at time t.
Let's imagine that each of them has access to the share of Fragmental Information $I_F(t)$ entrusted to them by the System, in the form of Fragmental Knowledge $K_F(t)$, and that they have the Encryption Keys, lent by the System itself, to decode the Fragment so as to fully understand its information content.

At this point, the Probability of Breaching the System will depend on:
- the amount of Information $I_F(t)$, in the form of Fragmental Knowledge $K_F(t)$, that each of them has;
- the random coherence of the individual shared Fragments: if N_{bad_agent} External Agents has "nearby" Fragments within the granular I_G Distribution, it is clear that the reconstruction of the original Information will be more likely than in the situation where External Agents share completely opposite Fragments within the same Distribution;
- the complexity of the System, understood not only as the total Volume of I_G

Information, but also by the Depth of Progressive Decomposition achieved, which is equivalent to the total number of Fragments (fragments(t) = $f_x(t)$) generated by the System;
- the degree of Entropy H(X) achieved by the System in its terminal Fragmentation.

At this point, the use of three mathematical tools allows us to demonstrate the Validity of the Hypothesis of the Theorem.

Shannon's Formula, as we saw in Chapter 11, explores the system's uncertainty in identifying the correct configuration of the terminal Fragments during the reconstruction, providing a measure of the entropy H(X) achieved by the I_G Information during the granulation process.

Bernoulli's Formula helps us in the Fragment assignment phase and allows us to calculate the probability that a Fragment will be randomly assigned to a Colluding External Agent, being a probability statistic in which an event can have only two outcomes (in our case, assigned to the Colluding External Agent or to the Honest External Agent).

Kullback-Leibler divergence measures the statistical "distance" between two probability distributions: in our case, we can use it to calculate the divergence between the critical Bernoulli distribution and the actual Bernoulli distribution. This statistical distance is a central point of the Theorem: that is, it evaluates when a specific Colluding Distribution could become a problem if it attacks the System, by comparing the Actual Colluding Distribution found in the System and the one considered Critical, that is, the Theoretical Distribution with a Critical Threshold sufficient to violate the System and corrupt it.

We therefore have:

Shannon Formula

$$H(X) = -K \, \Sigma_{i=1}{}^{N} \, p_i \log_2 p_i,$$

described in Chapter 10,

Bernoulli's Formula

$$P(Y=1) = q$$

$P(Y=0) = 1 - q$

which measures
event 1 = a fragment is assigned to a Colluding External Agent.
event 0 = a fragment is assigned to an Honest External Agent.
The Kullback-Leibler divergence for discrete cases is

$D(P\|Q) = H(P, Q) - H(P)$

where two Entropies are compared to each other, the Cross Entropy and the real Entropy of System P, that is, the Shannon Entropy which we have already seen and calculated in chapter 11.

Real Entropy $H = -\Sigma_x P(x) \log P(x)$

measures the intrinsic Uncertainty of the Real Distribution P, thus offering a measure of the Disorder of the Data (i.e. how “unpredictable” they are);

Cross Entropy $H(P, Q) = -\Sigma_x P(x) \log Q(x)$

It is a cross-measure, which measures the resulting Entropy if the events of the Real Distribution P are encoded using the approximate Model given by the Q Distribution.
If, for example, we wanted to measure tomorrow's weather, and we know that the Probability of Sunny is 70%, Cloudy is 20%, and Rain is 10%,
while our Estimate is 40% Sunny, 40% Cloudy, 20% Rain,
we would have
the Real Discrete Distribution 0.70, 0.20, 0.10 (A)
against
the Approximate Distribution 0.40, 0.40, 0.20 (B).
Cross Entropy is the comparison between these two distributions, obtained by measuring what happens to the Shannon Entropy of the Real Distribution (A) if

I use the data from the Approximate Distribution (B). The result, expressed in bits, tells us how inefficient the Approximation is, that is, how many more bits of information I am "wasting" because the Q Model (which corresponds to B in the Weather example) does not adhere perfectly to the P Model (A in the example).
In our case, we will have

$D(\tau \| q) = H(\tau, q) - H(\tau)$, which by replacing becomes

$$D(\tau \| q)=\tau \log (\tau / q) + (1 - \tau) \log ((1 - \tau)/ (1 - q)) \qquad \textbf{(68)}$$

where

τ, it is the minimum fraction of coherent information needed to attack effectively;

q, is the actual share of colluders .

Said differently $D(\tau \| q)$ measures how unlikely it is that an entity with a collusive share q will behave as if it had a collusive share τ.

For example, if q = 10% and the threshold required for Collusion Success is 30%,
that is, $q = 0.10$ and $\tau = 0.30$,
the Kullback-Leibler divergence measures how statistically distant 10% is from 30%.

In the proof, we introduced the value τ, the Critical Collusion Threshold. This value constitutes the minimum fraction of global information that colluders must control to breach the system effectively and consistently. Put another way, if colluders control less than τ, they do not have enough information to cause problems. However, if they control an amount of information with assigned $I_{F\_s} \geq \tau$, they can conduct an effective attack.

Let us assume that $q, \tau \in \mathbb{R}$ and $q, \tau \in (0, 1)$, and $n, c \in \mathbb{N}$

c = the number of Fragments controlled by the Colluding Agents
n = the total number of Fragments;
then the share controlled by the Colluding External Agents will be c/n.
Collusion will only be possible when $c/n \geq \tau$.
τ is a Model Parameter and can be estimated in various ways, but it is a parameter that should be provided.
Technically, it could be assumed by the Implementation in a "conservative" manner, but the most effective solution is for it to be estimated conceptually by analyzing how much minimal information is needed to reconstruct relationships, avoid contradictions, and violate the system's invariants. It is clear that this is difficult to calculate in an Implementation because it depends on the Nature of the Data we are dealing with, which is assumed to be abstract in this document.
Theoretically, it could also be estimated experimentally through Attack trials.
If we assume through analysis and simulations that at least 35% of the useful information is needed to violate the system, τ will be equal to 0.35, meaning that those who control less than 35% of the information will, on average, fail to violate the system.
As the system grows, the information I_G grows, and therefore the number of fragments increases, the entropy H(n) increases, and K_F decreases.
It follows that an increasingly larger amount of information is needed to reconstruct the system, thus requiring more colluders.
Formally,

$\tau = f_\tau(H, R, C)$ **(69)**
where we have
- H = System entropy;

- R = Structure of relations;
- C = System invariants.

That is, the Critical Threshold τ is a function $f\tau$ of the informational and structural complexity of the system, therefore of H, R, and C.

In the proof, we will use Bernoulli's Binomial Formula, which considers many independent Bernoulli trials.

Formally,

$P(X=k) = (n; k)\ q^k\ (1-q)^{\ n-k}$

where:

- n, the total number of Fragments;
- k, the number of fragments managed by the colluders;
- q, the probability that a single fragment goes to a colluder;
- P(X=k) the probability that k fragments are assigned to the Colluding External Agents.

To recap, with the simple Bernoulli's Formula, the probability that a single fragment goes to a colluder is established; Bernoulli's Binomial Formula calculates the probability that out of all the fragments (i.e., in a distribution), a quantity k goes to the colluders.

In the demonstration we will use Stirling's Formula that we have already seen in the Discussion on Entropy in chapter 11, namely

$n! \approx \sqrt{(2\pi n)} * (n/ e)^n$, and which constitutes an approximation when calculating the factorials of very large numbers.

Proof

The statement of the Emergent Security Theorem is:

$$q\bar{\omega} < \tau \implies \lim_{n \to \infty} P\left[\Sigma^{n}_{i=1} Y_i \omega (d_i) \geq \tau n\right] = 0 \quad \textbf{(70)}$$

The more the System is adopted, the more useful knowledge is dispersed, and the more unlikely, costly, and technically inconvenient collusion becomes.
Let's start with the idea of having a certain number of terminal Fragments n.
For simplicity's sake, let's assume that they are all equally probable.
If we want to recombine them all, the total number of all possible combinations corresponds to n!, and these are Permutations, that is, taking each individual Fragment only once, and taking into account the importance of the order in which they are distributed. In fact, the Fragments contain information and cannot be distributed in a random order.
As we have already seen in Chapter 11, the formula for calculating each individual Probability of a specific Permutation will be p = 1/ n!.

Let Shannon Entropy be $H = -\Sigma_i \; p_i \log_2 (p_i)$,

let's calculate the Entropy of our distribution of terminal Fragments n obtained from the System, and substitute the terms

$$H(n) = -\Sigma^{n!}_{i=1} \; 1/ n! \log_2 (1/ n!),$$

simplifying the same terms, we will have

$$H(n) = -n! \cdot 1/ n! \log_2 (1/ n!),$$

simplifying further we will have

$$n! \cdot 1/ n! = 1,$$

therefore,

$H(n) = -\log_2 (1/ n!)$,

since in logarithms the property $\log(a^{-1}) = -\log(a)$ holds

$a^{-1} = 1/ a$, we will still have

$-\log_2 (1/ a) = \log_2 (a)$, and let's set $a = n!$

we will have that the above relation can be reduced to

$H(n) = \log_2 (n!)$.

But as the fragments increase, the possible combinations also increase, so the reconstruction uncertainty ultimately increases. We've already seen this in Chapter 11.
Let's construct the limit, and then we have $n \to \infty \;\Rightarrow\; H(n) \to +\infty$

$\lim_{n \to \infty} H(n) = \lim_{n \to \infty} \log_2 (n!) = +\infty$.

Let's move on to using Bernoulli's distribution to randomly assign each individual Terminal Fragment (simple formula) and calculating the probability that, of all the n distributed Fragments, some will end up in the hands of the Colluding External Agents (binomial formula).
For the purposes of proving the theorem, we use Bernoulli's binomial distribution to calculate the probability that a number k of Fragments will be assigned to the Colluding External Agents. The value of k subsequently determines the reduction in the entropy of the system.
So, let's first calculate the probability that, by chance, each individual Terminal Fragment will end up in the hands of a Colluding External Agent instead of an Honest External Agent.
When the system assigns this fragment, only two things can happen: either the fragment goes to a Colluding External Agent or it goes to an Honest External

Agent.
There are no other possibilities.
Let's introduce the variable Y, which can take only two values.
Let's assume that
Y = 1, if the Fragment randomly goes to the Colluding External Agent,
Y = 0, if the Fragment randomly goes to the Honest External Agent.

Let q be the probability that a single terminal Fragment ends up in the hands of a Colluding External Agent, and we have
$P(Y=1) = q$.
Since there are only two possibilities, the opposite probability is
$P(Y=0) = 1 - q$.
At this point, we use Bernoulli's Binomial Formula to calculate the probability of multiple Fragments (potentially all):
$Y_1, Y_2, \ldots, Y_n$
by adding them together, we have
$X = Y_1 + Y_2 + \cdots + Y_n$
where X represents a random variable that counts the total number of Fragments assigned to the Colluding External Agents.
Suppose we have the following situation (1 = Colluder, 0 = Honest):
$Y_1 = 1$
$Y_2 = 0$
$Y_3 = 1$
$Y_4 = 0$
$Y_5 = 1$
Then we have:
$X=1 + 0 + 1 + 0 + 1 = 3$
which means that out of 5 Fragments, 3 ended up in the hands of the Colluding External Agents.

The probability that exactly k terminal Fragments go to the Colluders is calculated with the following formula:

$P(X=k) = (n ; k)\, q^k (1-q)^{n-k}$.

We must pay attention to the meaning of X and k.

X is the random variable that determines the number of fragments assigned to the Colluder;

k corresponds to that specific number.

In fact $k \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n\}$,

therefore

X = random variable that counts the fragments to the Colluders.

k = the particular value assumed by the variable X.

Let's return to the proof.

We stated that the System Entropy corresponded to

$H(n) = \log_2(n!)$, which corresponds to the Uncertainty of the Entire System.

We should subtract a portion of these Fragments from the System Entropy calculation, because they will end up in the hands of the Colluders, reducing the Entropy.

The System Entropy is reduced because the Colluding External Agents, receiving the Fragments, will know their contents, therefore a portion of the I_G Information will no longer be unknown and will no longer contribute to determining Uncertainty (one thing known is certain).

The Number of fragments still uncertain and unknown will be equal to n - k, those that ended up in the hands of the Honest External Agents.

The residual entropy will be

$Hres(k) = \log_2((n - k)!)$, counting only the unknown Fragments that ended up in the hands of Honest External Agents.

The entropy reduction will therefore be

$\Delta H(k) = H(n) - Hres(k)$, substituting the values H(n) and $H_{res}(k)$ we will have

$\Delta H(k) = \log_2(n!) - \log_2((n - k)!)$.

Exploiting the property of logarithms that $\log(a) - \log(b) = \log(a/b)$, substituting the terms we will have

$\Delta H(k) = \log_2(n! / (n - k)!)$.

In summary,

- If $k = 0$, the colluders know nothing,
- If k increases, uncertainty decreases,
- If $k = n$, uncertainty becomes zero, because everything is known by the Colluding Agents.

So, up to this point,

the application of the Shannon Formula allows us to measure the Total Entropy of the Information I_G, the Bernoulli Formulas measure the Probability of obtaining k, and k determines how much Entropy is "eliminated." $\Delta H(k)$ provides the value of the reduction in the Entropy of I_G, and τ is the correction factor for n, which defines the minimum number of Fragments that must be controlled to reduce the Entropy of I_G to the point of allowing a potential attack capable of violating System Security.

If the reduction in System Entropy becomes large enough to exceed a Threshold Value, Collusion goes from being an unlikely event to a "dangerous" event. This occurs when $k \geq \tau \cdot n$.

The probability that collusion leads to success, thus deteriorating the system's security, is therefore given by

$P(X \geq \tau \cdot n)$

Summary table:

$c = k$ = number of Fragments assigned to the Colluders

$c/ n = k/ n$

$k/ n \geq \tau$

$k \geq \tau \cdot n$

$k\tau = \lceil \tau \cdot n \rceil$

$k \geq k\tau$

Starting from the Binomial Formula, we calculate the total probability of success of a collusion, which will therefore be given by the sum of all the probabilities of the cases in which the Colluders receive enough Fragments to reach or exceed the Critical Threshold.

The probability of success of the collusion will therefore be:
$P_{coll}(n) = P(X \geq k\,\tau)$, that is,

$$P_{coll}(n) = \Sigma^{n}{}_{k=k\_\tau} \; P(X = k)$$

with X (= Y1 + Y2 + ⋯ + Yn) representing the variable referring to the total number of Fragments assigned to the Colluding External Agents and k a specific number, we will have

$$k_{\tau} = \lceil \tau \cdot n \rceil = \text{Critical Threshold value.} \qquad \textbf{(71)}$$

Substituting we will have,

$$P_{coll}(n) = \Sigma^{n}{}_{k=\lceil \tau \cdot n \rceil} \; (n\,;k)\; q^{k}\,(1-q)^{n-k}. \qquad \textbf{(72)}$$

The formula to be proved was:

$$q\bar{\omega} < \tau \implies \lim_{n \to \infty} P\,[\Sigma^{n}{}_{i=1}\; Y_i \omega\,(d_i) \geq \tau n\,] = 0$$

$$q < \tau \implies \lim_{n \to \infty} P_{coll}(n) = 0. \qquad \textbf{(73)}$$

with

$X \sim \text{Bin}(n, q)$,

that is, X is the variable of the Binomial, which counts how many fragments end up with the Colluders,
n, the total number of Fragments,
q, the Probability of success of a single Bernoulli Trial, i.e., $q = P(Y_i = 1)$, which is the Probability that a single Fragment will be randomly assigned to the Colluder.
The expected value is: $E[X] = n \cdot q$,

so the Colluders receive on average $n \cdot q$ fragments.
The minimum threshold is $k = k_\tau = \lceil \tau \cdot n \rceil$,
but if
$q < \tau$
then,
$n \cdot q < \tau \cdot n$
that is, the average number of fragments received by the Colluders is less than the threshold required to violate the System.
Therefore,
X/n will be the fraction of the total Fragments that end up with the Colluders.
Collusion is therefore successful if $X/n \geq \tau$,
so substituting we have
$P_{coll}(n) = P(X/n \geq \tau)$.
By the law of large numbers,
if $n \to \infty$
we have
$X/n \to q$,
that is, if the number of Fragments grows infinitely, the effective and real fraction of Fragments assigned to the Colluders increasingly approaches Probability q, which is the probability that a single fragment is assigned to a Colluder: therefore, the observed fraction converges to the actual value q.
But if the real fraction approaches q, and q is smaller than τ, then it becomes increasingly unlikely that
$X/n \geq \tau$, so substituting we have:

$q < \tau \implies \lim_{n \to \infty} P(X/n \geq \tau) = \lim_{n \to \infty} P_{coll}(n) = 0$, therefore

$q < \tau \implies \lim_{n \to \infty} P(X \geq \lceil \tau \cdot n \rceil) = 0$, by replacing we will have

$q < \tau \implies \lim_{n \to \infty} \Sigma^{n}_{k=\lceil \tau \cdot n \rceil} (n ; k) \; q^{k} (1 - q)^{n-k} = 0$,

That is, if the actual share of Colluders q is less than the minimum share τ, then by increasing the number of fragments, it becomes increasingly less likely that the Colluders will randomly receive enough fragments to exceed the critical threshold.
Let's return to Kullback-Leibler divergence, which we saw at the beginning of this chapter:

$D(\tau\|q)=\tau \log (\tau/ q) + (1 - \tau) \log ((1 - \tau)/(1 - q))$

D measures how far the actual situation q is from the critical level τ, that is, from the situation in which the Critical Threshold level for assigning Fragments to Colluders is actually reached, or
it measures how unlikely it is that a binomial variable with actual probability q will produce an observed frequency equal to at least τ.
For a binomial, the Chernoff Inequality holds, which is its classic application. When the number of independent trials (in this document, corresponding to Fragments) is very high, calculating the Binomial is complex.

The Chernoff Inequality allows us to obtain an upper and a lower limit with respect to the mean of the Probability of the Binomial, thus obtaining an Interval within which the expected Probability of the Binomial falls:

upper bound $P(X \geq (1 + \delta)\mu) \leq e^{-\delta^2 \mu / 3}$

lower bound $P(X \leq (1 - \delta)\mu) \leq e^{-\delta^2 \mu / 2}$

That is, the further one moves from the mean (and therefore δ is large) or the more one increases the number of trials (which correspond to μ), the more the probability of an anomalous event (with respect to the expected value from the binomial) collapses towards zero.
Applied to a binomial, we can therefore calculate the probability of departing from the expected interval: $P(|X - \mu| \geq \delta\mu) \leq 2e^{-\delta^2 \mu / 3}$.

Applying the Chernoff bound to the upper tail of the binomial distribution we obtain:

$e^{-nD(\tau\|q)}$

therefore

$P(X/\,n \geq \tau) \leq e^{-nD(\tau\|q)}$ **(74)**

if $\tau > q$,

since $P_{coll}(n) = P\,(X/\,n \geq \tau)$,

we can also write

$P_{coll}(n) \leq e^{-nD(\tau\|q)}$, **(75)**

that is, the probability that the colluders exceed the critical threshold (= Pcoll (n) = P (X/ n ≥ τ)) drops exponentially the greater the number of trials (i.e. Fragments) n, and the greater the distance between q and τ calculated with the Kullback-Leibler divergence.

Let's say $\Gamma = D(\tau\|q)$ so as to simplify writing the formula.

if $\Gamma > 0$

then $P_{coll}(n) \leq e^{-n\Gamma}$

as n increases, the exponent becomes more and more negative:

assuming $\Gamma = 0.1$

$n = 10 \quad \rightarrow e^{-1}$

$n = 100 \quad \rightarrow e^{-10}$

$n = 10000 \rightarrow e^{-100}$

The value will tend exponentially to 0.
Since, $\Gamma > 0$, we have:

$\lim_{n \rightarrow \infty} e^{-n\Gamma} = 0.$

Since $0 \leq P_{coll}(n) \leq e^{-n\Gamma}$

and for $e^{-n\Gamma}$ holds $\lim_{n \rightarrow \infty} e^{-n\Gamma} = 0,$

therefore $P_{coll}(n) \rightarrow 0$ for $n \rightarrow \infty$, concluding

$q < \tau \implies \lim_{n \rightarrow \infty} P_{coll}(n) = 0.$

In reality, not all Fragments have the same Weight.
Some Fragments are more useful than others because they are better connected.
If two External Agents receive two Fragments that are each a continuation of the other, it is obvious that for each of the two Agents, the Weight of the related I_F information received is significantly higher than if both received two completely unrelated Fragments.
Let's try to introduce a Weight measure, which we will denote as

$Y_i\omega(d_i)$, where

$Y_i = 1$, if the Fragment is assigned to the Colluder,

$Y_i = 0$, if it is assigned to an honest one,

instead

$\omega(d_i)$ measures how informationally useful that Fragment is compared to the others.

In detail

$\omega(d_i) \in [0, 1]$,

if $\omega(d_i) \approx 1$, The Fragment is very useful for the purposes of information reconstruction,

if $\omega(d_i) \approx 0$, the Fragment is completely disconnected from the rest,

In short:
The Fragment only counts if it goes to a colluder, and it counts based on the weight of its informational utility.

$$\bar{\omega} = 1/n \ \Sigma^{n}_{i=1} \ \omega(d_i) \qquad \textbf{(76)}$$

where $\bar{\omega}$ constitutes the Average Value of the Information Utility of the Fragments.

Therefore $q\bar{\omega}$,

represents the effective collusive share, i.e., the corrected share of the weight of the Average Value of the Information Utility of the Fragments.
Therefore, if

$q\bar{\omega} < \tau$,

the Expected Value remains below the Critical Threshold, if

$$E\left[\Sigma^{n}_{i=1} Y_i\omega(d_i)\right] = q\,\Sigma^{n}_{i=1}\omega(d_i) = nq\bar{\omega}, \qquad \textbf{(77)}$$

where E denotes the expected value, i.e. the theoretical average amount of useful information controlled by the colluders.
In conclusion, the Formula

$$q < \tau \implies \lim_{n\to\infty} P_{coll}(n) = 0,$$

which is equivalent to explicitly

$$q < \tau \implies \lim_{n\to\infty} P(X \geq \lceil\tau\cdot n\rceil) = 0,$$

By adding the value $\omega(d_i)$ for each Y_i to X, and substituting,
we will finally have:

$$q\bar{\omega} < \tau \implies \lim_{n\to\infty} P\left[\Sigma^{n}_{i=1} Y_i\omega(d_i) \geq \tau\cdot n\right] = 0,$$

for $0 \leq \omega(d_i) \leq 1$

which means:

<u>If the effective collusive share, weighted by the average informational utility of the fragments, remains below the critical threshold, then as the number of fragments grows, the probability that colluders obtain enough useful information to compromise the system tends to zero.</u>

******Decomposition Tables

with ρ_C=0.30, α=0.70, β=0.30, I_0 = 1M

N = 1 constant node

I_G	λ=1	λ=2	λ=5	λ=10
100k	4	7	17	34
200k	4	7	17	34
300k	4	7	17	34
400k	4	7	17	34
500k	4	7	17	34
600k	4	7	17	34
700k	4	7	17	34
800k	4	7	17	34
900k	4	7	17	34
1M	4	7	17	34

N = 2 constant nodes

I_G	λ=1	λ=2	λ=5	λ=10
100k	3	6	14	28
200k	3	6	14	28

I_G	λ=1	λ=2	λ=5	λ=10
300k	3	6	14	28
400k	3	6	14	28
500k	3	6	14	28
600k	3	6	14	28
700k	3	6	14	28
800k	3	6	14	28
900k	3	6	14	28
1M	3	6	14	28

N = 3 constant nodes

I_G	λ=1	λ=2	λ=5	λ=10
100k	3	5	12	24
200k	3	5	12	24
300k	3	5	12	24
400k	3	5	12	24
500k	3	5	12	24
600k	3	5	12	24
700k	3	5	12	24
800k	3	5	12	24
900k	3	5	12	24
1M	3	5	12	24

---

con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 100.000

I_G	Log factor
100k	1.000
200k	1.585
300k	2.000
400k	2.322
500k	2.585
600k	2.807
700k	3.000
800k	3.170
900k	3.322
1M	3.459

Case N = 1 constant nodes

I_G	Frammenti per nodo
100k	4
200k	6
300k	7
400k	8
500k	9
600k	10
700k	10
800k	11
900k	12
1M	12

Case N = 2 constant nodes

I_G	fragments	Total
100k	3	6
200k	5	10
300k	6	12
400k	7	14
500k	8	16
600k	8	16
700k	9	18
800k	9	18
900k	10	20
1M	10	20

Case N = 3 constant nodes

I_G	fragments	Total
100k	3	9
200k	5	15
300k	6	18
400k	7	21
500k	7	21

I_G	fragments	Total
600k	8	24
700k	9	27
800k	9	27
900k	10	30
1M	10	30

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con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 10.000

Case N = 1 constant nodes

I_G	fragments
100k	12
200k	15
300k	17
400k	18
500k	19
600k	20
700k	21
800k	22
900k	22
1M	23

Case N = 2 constant nodes

I_G	fragments	Total
100k	10	20
200k	12	24
300k	14	28
400k	15	30
500k	16	32
600k	17	34
700k	17	34
800k	18	36
900k	18	36
1M	19	38

Case N = 3 constant nodes

I_G	fragments	Total
100k	8	24
200k	11	33
300k	12	36
400k	13	39
500k	14	42
600k	15	45
700k	15	45
800k	16	48
900k	16	48
1M	17	51

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con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 100.000

Node	λ=1	λ=2	λ=5
1	12	24	58
2	10	48 total	96 total
3	10	57 total	144 total
5	9	80 total	195 total
10	7	140 total	580 total

END DOCUMENT

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